

Arjun Chavan

akchavan@umich.edu | linkedin.com/in/arjun-chavan | github.com/ArjunChavan2 | leetcode.com/u/arjunicus

EDUCATION

University of Michigan, Ann Arbor, MI

Expected Dec 2027

Bachelor of Engineering in Computer Science

Relevant Coursework: Data Structures and Algorithms, Computer Vision, Computer Architecture, Applied Linear Algebra, Discrete Mathematics; Machine Learning, Embedded Systems, and Autonomous Robotics (In Progress)

SKILLS

Languages: Python, C, C++, Java, SQL, JavaScript, TypeScript, HTML, CSS, Bash, LaTeX, Verilog, R

Frameworks and Tools: React, Vite, Node.js, REST APIs, Git, GitHub, Linux, Unix, VS Code, Figma, Agile, Scrum

AI and ML: OpenCV, PyTorch, TensorFlow, Prompt Engineering

Hardware: Circuit Design, Arduino, Raspberry Pi, Magnetic Induction, Soldering, FPGAs

EXPERIENCE

Tech Lead, Altheros Capital, Ann Arbor, MI

May 2026 – Present

- Co-led a 10-engineer fullstack team on Midwest Health, a healthcare patient-scheduling platform for an Altheros Capital portfolio company, planning sprints across the stack
- Reviewed pull requests and held merge authority for the team, gating all changes before production deployment
- Presented monthly demos of the Midwest Health and Nerd Perfect Media applications to a 60-person audience including an Altheros Capital partner and portfolio leadership
- Scoped a two-week MVP timeline and two-developer staffing plan for Nerd Perfect Media, recommendation adopted
- Led a schema audit, facilitating cross-team discussion to align frontend and backend on a shared data format

Full Stack Developer Intern, Altheros Capital, Ann Arbor, MI

Feb 2026 – Aug 2026

- Built landing pages and UI components in React, JavaScript, and CSS for the Midwest Health and Nerd Perfect Media applications, matching Figma specifications
- Collaborated in Agile and Scrum sprint planning, translating user stories into actionable UI tasks
- Migrated build tooling from deprecated react-scripts to Vite, restoring CI pipeline stability
- Authored the team deployment runbook covering release packaging, secure upload, and rollback procedures

Machine Learning Research Assistant, UMass Lowell (PeARL Lab), Lowell, MA

May 2023 – Aug 2023

- Built machine learning models to optimize robotic arm pathfinding and manipulation, improving task success rates
- Automated data collection pipelines in Python, reducing preparation time for the research team
- Deployed control scripts that increased the success rate of robotic pick-and-place operations

Machine Learning Engineer, MIT Lincoln Laboratory (Beaver Works), Cambridge, MA

July 2023 – Aug 2023

- Implemented autonomous navigation using OpenCV to track pool lane lines in real time
- Processed AprilTag detection data to compute relative 3D position and heading between robotic units
- Tuned PID controllers to stabilize robot orientation, producing smoother and more consistent trajectory tracking

PROJECTS

Gamified Financial Dashboard, React, Expo, Node.js, TypeScript, CMU TartanHacks

Feb 2026

- Created a gamified personal finance dashboard with an AI-driven virtual pet system and real-time quest overlays
- Developed a TypeScript onboarding engine parsing the Capital One REST API to surface AI-curated financial insights
- Secured 1st Place in the Visa Track, presenting a scalable fintech solution to industry executives

Underwater Inspection ROV, C++, Arduino, 4-Person Team Design Project

Jan 2025 – Apr 2025

- Built the C++ Arduino control system for a 4-thruster inspection ROV, abstracting surge, heave, and yaw into five buttons and a toggle switch
- Added a latching cruise control that locked forward thrust for long transits, improving measured forward speed by 1.5x in lab testing
- Briefed engineering stakeholders on hardware and software integration through a live technical demo

Facial Emotion Recognition, Python, TensorFlow, OpenCV

Apr 2024

- Trained a real-time emotion classifier using facial landmark detection and a deep neural network
- Achieved 60% validated prediction accuracy, outperforming baseline emotion classifiers on the test set